Theoretical Research Report: EABC-Proof-Theory (Version 19)

Research Collaboration, Thomas Hoffbauer - Gemini

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Abstract

This manuscript finalizes the structural, algebraic, and measure-theoretic foundations of the EABC-Proof-Theory research program. We establish the orthogonal Pythagorean ANOVA-decomposition of the Hurwitz-Hoffbauer defect, derive its precise relation to unnormalized window-integrated two-point functionals, prove the topological classification of frequency sets as $\mathbf{\Pi}^0_3$-complete, and analyze the asymptotic behavior of the 8-state isotropic reward-controlled Markov kernel.

1 Pythagorean Orthogonal Decomposition and Variance Splitting

Let $\eta_{\min} := (1,1,1,1)$ denote the minimal EABC reference pattern, and let $P = (h_E, h_A, h_B, h_C)$ be the observed Hoffbauer bracket. Define the deviation vector by

$$\delta := P - \eta_{\min} = (\delta_E, \delta_A, \delta_B, \delta_C), \qquad \delta_X := h_X - 1,$$

and its total cumulative excess by

$$K := \langle \delta, \mathbf{1}_4 \rangle = \sum_{X \in \{E,A,B,C\}} \delta_X, \qquad \mathbf{1}_4 := (1,1,1,1).$$

Since $\|\mathbf{1}_4\|_2^2 = 4$, the real vector space $\mathbb{R}^4$ decomposes orthogonally into the radial line and its orthogonal complement:

$$\mathbb{R}^4 = \operatorname{span}\{\mathbf{1}_4\} \overset{\perp}{\oplus} \mathbf{1}_4^{\perp}, \qquad \delta = \underbrace{\frac{K}{4}\mathbf{1}_4}_{\delta_\parallel} + \underbrace{\left(\delta - \frac{K}{4}\mathbf{1}_4\right)}_{\delta_\perp}.$$

By construction, $\langle \delta_\perp, \mathbf{1}_4 \rangle = 0$.

Define the unminimized direct defect by

$$D^0_{\mathrm{HH}}(P) := \|\delta\|_2^2 = \sum_{X \in \{E,A,B,C\}} \delta_X^2.$$

The Pythagorean theorem then yields the exact variance decomposition:

$$\boxed{D^0_{\mathrm{HH}}(P) = \underbrace{\frac{K^2}{4}}_{\text{radial component of total count excess}} + \underbrace{D^{\perp}_{\mathrm{HH}}(P)}_{\text{internal EABC asymmetry (transversal)}}}$$

where the absolute transversal shape defect is given by

$$\begin{aligned} D^{\perp}_{\mathrm{HH}}(P) := \|\delta_\perp\|_2^2 &= \sum_{X \in \{E,A,B,C\}} \left(\delta_X - \frac{K}{4}\right)^2 \\ &= \sum_{X \in \{E,A,B,C\}} \delta_X^2 - \frac{1}{4}\left(\sum_{X \in \{E,A,B,C\}} \delta_X\right)^2 = 4\ \mathrm{Var}_{\mathrm{pop}}(\delta_E, \delta_A, \delta_B, \delta_C). \end{aligned}$$

1.1 Representation as an Integrated Two-Point Functional

Let I be a finite window of EABC prime classes, and define

$$N := |I|, \qquad h_X := \#\{t \in I : c_t = X\}.$$

Then

$$\sum_{X\in\{E,A,B,C\}} h_X = N$$

and consequently

$$K = \sum_X (h_X - 1) = N - 4.$$

Using the centered simplex embedding

$$v(c_t) := e_{c_t} - \frac{1}{4}\mathbf{1}_4 \in \mathbf{1}_4^{\perp},$$

one obtains $\sum_{t\in I} v(c_t) = \delta_\perp$. Therefore,

$$\boxed{\begin{aligned} D^{\perp}_{\mathrm{HH}}(I) &= \left\| \sum_{t\in I} v(c_t) \right\|_2^2 \\ &= \sum_{r,s\in I} \langle v(c_r), v(c_s)\rangle \\ &= \sum_{X\in\{E,A,B,C\}} h_X^2 - \frac{N^2}{4}. \end{aligned}}$$

Thus $D^{\perp}_{\mathrm{HH}}$ operates strictly as an unnormalized, window-integrated algebraic two-point functional of the centered EABC simplex representation without requiring physical assumptions.

1.2 Range of the Relative Anisotropy Index

For $D^0_{\mathrm{HH}}(P) > 0$, define the relative anisotropy index

$$R_\perp(P) := \frac{D^{\perp}_{\mathrm{HH}}(P)}{D^0_{\mathrm{HH}}(P)} = 1 - \frac{K^2}{4D^0_{\mathrm{HH}}(P)}.$$

It is invariant under nonzero rescalings $\delta \mapsto \lambda\delta$. Its admissible range depends on the domain:

1. **General defect space.** In $\mathbb{R}^4$ (including $\mathbb{Z}^4$), Cauchy–Schwarz yields $0 \le R_\perp \le 1$, with

$$R_\perp = 0 \iff \delta \in \mathrm{span}\{\mathbf{1}_4\}, \qquad R_\perp = 1 \iff K = 0,\ \delta \neq 0.$$

2. **Positive excess cone $\delta_X \ge 0$.** The nonnegativities and Cauchy–Schwarz yield the dual enclosure $D^0_{\mathrm{HH}}(P) \le K^2 \le 4D^0_{\mathrm{HH}}(P)$, and therefore

$$\boxed{0 \le R_\perp \le \frac{3}{4}.}$$

The lower bound corresponds to purely radial growth ($4 \mid K$), while the upper bound $R_\perp = \frac{3}{4}$ occurs if and only if the excess concentrates in a single channel ($\delta = (K, 0, 0, 0)$ up to permutation).

2 Markov-Kernel Analysis and Entropy Limits

For fixed $T_{\text{eff}} > 0$, define

$$\beta_P := \frac{P^2}{T_{\text{eff}}}, \qquad E_{ij}(P) := -P^2 K_{ij}, \qquad K_{ii} = 0, \qquad K_{ij} = 1 \quad (i \neq j).$$

The associated Gibbs kernel on the 8-state space $S_8 := C \times \{+, -\}$ is

$$T_{ij}(P) = \frac{\exp(\beta_P K_{ij})}{\sum_{k=1}^{8} \exp(\beta_P K_{ik})}.$$

As $P \to \infty$ ($\beta_P \to \infty$), one has

$$T(P) \longrightarrow T_\infty := \frac{1}{7}(J_8 - I_8).$$

2.1 Entropy and Information Limits

In the high-temperature limit $\beta_P \to 0$, the row entropy is $H_{\text{high}} = \log_2 8 = 3$ bits. In the low-effective-temperature limit $\beta_P \to \infty$, the row entropy is

$$H_\infty = \log_2 7 \approx 2.80735 \text{ bits}, \qquad \Delta H = \log_2 \frac{8}{7} \approx 0.192645 \text{ bits}.$$

Under the stationary distribution of the limiting chain, the mutual information is

$$I(X_n; X_{n+1}) = \log_2 \frac{8}{7},$$

recording exactly the exclusion of the current state from the seven admissible successors.

2.2 Spectral Characterization

The spectrum of T_∞ is $\text{spec}(T_\infty) = \{1, (-1/7)^{[7]}\}$. With $\Pi := \frac{1}{8} J_8$, the powers satisfy

$$T_\infty^n = \Pi + \left(-\frac{1}{7}\right)^n (I_8 - \Pi),$$

ensuring geometric convergence of every initial distribution to the uniform stationary distribution.

3 Epistemic Status Summary

Component	**Status**
Pythagorean ANOVA-Decomposition ($D^0_{\text{HH}} = \frac{K^2}{4} + D^\perp_{\text{HH}}$)	[T] Proved
Integrated Two-Point Functional Representation	[T] Proved
Anisotropy Index Bounds ($0 \leq R_\perp \leq \frac{3}{4}$ in positive cone)	[T] Proved
Frequency Sets Topological Classification ($\mathbf{\Pi}^0_3$-completeness)	[T] Proved
Markov Grenzkernel $T_\infty = \frac{1}{7}(J_8 - I_8)$	[T] Proved
Physical Crystallization / Zero-Entropy Collapse	[F] Refuted (Kernel has $\log_2 7$ entropy)